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ABSTRACT

This project tested which Attention mechanism that can be implemented into a Neural Machine Translation (NMT) model is better suited for the translation of low-resource datasets using a pretrained model with a small dataset. Low-resource languages often lack sufficient training data to be able to accurately translate. This poses a problem as people native to the language have fewer accessible resources and the languages have a higher chance of going extinct.

The research utilized pre-trained models from an open source repository, Hugging Face Hub, that used small English to German datasets to mimic low-resource languages. Three different mechanisms were implemented to three NMT models and fine-tuned before being tested using installed programs from OpenNMT, an open source repository. Scaled-dot product Attention, Multi-head Attention, and Cross Attention were implemented onto a Transformer architecture. The hypothesis was that the NMT model will be the most accurate with the implementation of the Transformer with the Multi-head attention mechanism because it will additionally incorporate the Scaled Dot mechanism but not be exaggerated in the number of mechanisms working at once. This hypothesis was based on the rationale that in regards to small datasets, having too many mechanisms working in the model would lead to a contextually inaccurate translation.

The hypothesis did not align with the results of the experiment. According to the Bilingual Evaluation Understudy (BLEU) metric and Perplexity metric used, the tests that provided the highest BLEU score used the Cross Attention mechanism. This is relevant to improving translation for low-resource languages as it improves the model's response to

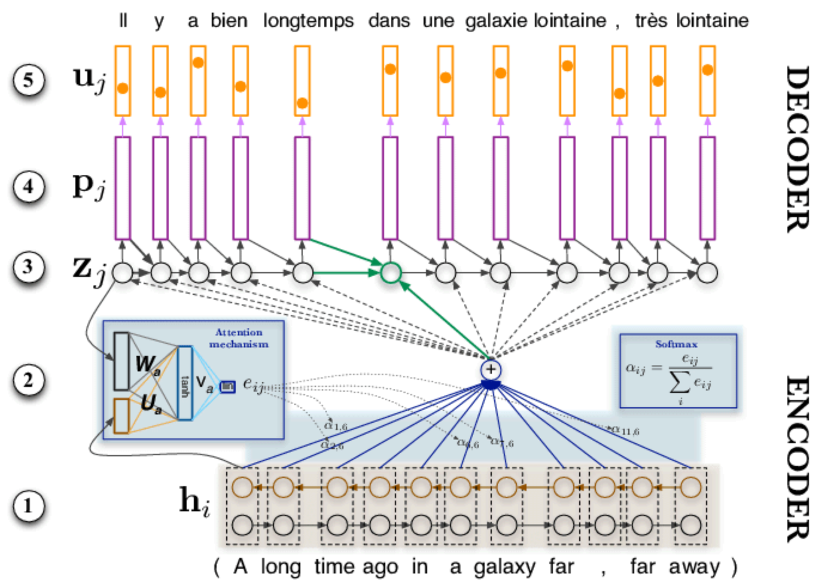
input without further training. The model and experiment were made cost-accessible, however, it would be beneficial to consider in the future how the Cross Attention mechanism would perform for true low-resource language pairs.

INTRODUCTION

Neural Machine Translation Models

Language barriers act as a hurdle in the advancement of human communication. Neural Machine Translation Models (NMTs) aim to solve this issue using an approach to Machine Translation that implements artificial deep neural networks, mimicking the neurons of the human brain, in order to predict the likelihood of a sequence of words (Bahdanau et al., 2014). These models are a type of software, made of lines of code written by researchers that serve as instructions for the computer on how to interpret inputs by a user. These models typically do not have a graphical representation or visual interface for the user unless programmed by the creator.

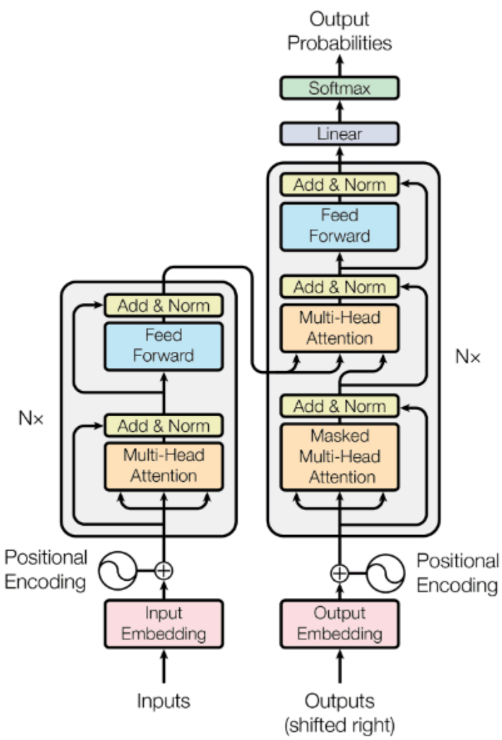
Given the fluidity of language, machine translation is a challenging artificial intelligence task. These models provide an accurate translation that instantaneously takes into account linguistic inconsistencies as they are able to learn and build off prior knowledge. NMTs replace statistical approaches to translation. They employ a system of tokenization and vectorization in the encoding process, turning the input text into vectors that are passed through a transformer. The transformer is a type of architecture that modifies the way a model encodes and decodes the input. The decoder then reconstructs the vector back into tokens, this time in the target language (see Figure 1).



Source: https://www.researchgate.net/figure/Architecture-of-the-NMT-system-equipped-with-an-attention-mechanism_fig1_308152598

(Figure 1)

Training and Transformers



Source: https://www.researchgate.net/figure/Architecture-of-the-NMT-system-equipped-with-an-attention-mechanism_fig1_308152598

Figure 2.

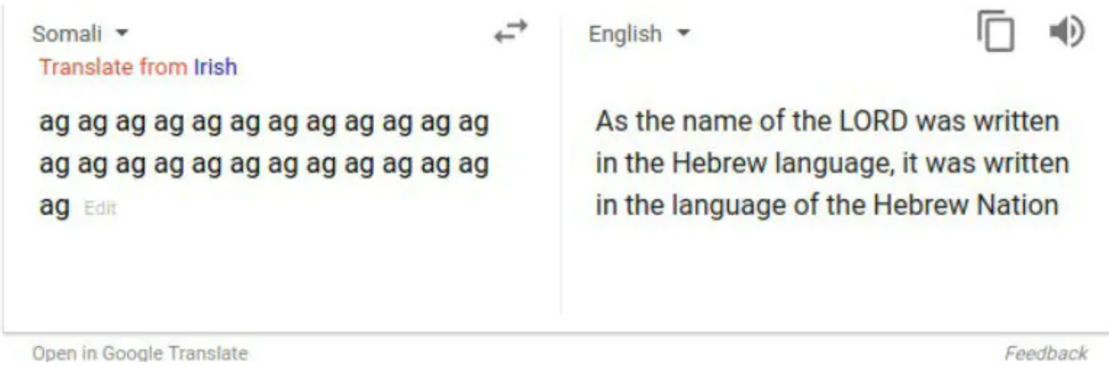
NMT models must be trained on a dataset for the specified language pair, the size and source of the data affecting the quality of the translation. In this case an English to German dataset was used to train the model. NMT models are ever-improving and can be further trained or fine-tuned with additional datasets or algorithms. An Attention mechanism is a mathematical concept that helps the model capture long-range dependencies and improve the overall quality of translations, especially in the presence of complex sentence structures or when translating between languages with different word orders. The Attention mechanism makes up the Transformer architecture that can be applied to a model (see Figure. 2).

Low-source Languages

Neural Translation Models (NMTs) allow for translation between languages using generative artificial intelligence. These models provide an accurate translation that instantaneously takes into account linguistic inconsistencies. This includes idiomatic expressions that cannot be directly translated and must take context into account. For example, "It's raining cats and dogs" cannot be directly translated into spanish, "Esta lloviendo gatos y perros" as the phrase means that it is raining hard, and not that cats and dogs are raining from the sky.

In fine-tuning these models, applying advanced techniques such as attention mechanisms or algorithms, these models can become even more specific and errorless. Attention mechanisms aim to interpret text in a similar way that a human would. For example, a previous translation model interpreting the phrase, "He liked the book because it was funny" would not understand that the word "it" refers to "the book" previously mentioned in the sentence. Attention allows for these types of contextual passages to be interpreted especially in long range dependencies, or longer texts. In improving these models which mimic human cognitive functions, multilingual communication will be facilitated in order to ensure that language does not act as a barrier in expressing ideas.

Additionally, Attention and the Transformer aim to improve models without the need for additional training. This would mean that models would become better at translating texts no matter the size of the dataset the original model was trained on. This is important as it allows for the possibility of languages with small datasets available, or low-resource languages such as extinct or dying languages, to be translated at higher standards. Within the large scale, this could eventually lead to reviving many extinct or dying languages.



Source: https://www.gadgets360.com/apps/news/google-translate-religious-texts-doomsday-prophecies-end-of-world-1888133

Figure 3.

Above in figure 3 is an example of issues in translating low-resource languages. In this case, the Google Neural Machine Translation model was trained on mostly religious texts for the language of Somali, which led to errors in translation. The model translated random strings of letters in Somali into religious sentences.

With low-resource datasets there is also the risk of the model overfitting if it is over-trained or there are too many mechanisms at work. Overfitting means that the model has trained too much on a small dataset so that it learns the errors and exceptions as new concepts. This, in turn, leads to the model not being able to apply the concepts it knows to the input text.

PROBLEM

Many indigenous and underrepresented languages are considered low-resource in their datasets and are difficult to translate. For example, languages like Vietnamese, Swahili, Hindi, Thai, Urdu, Bengali, Hawai'ian have a small digital presence and a low amount of datasets. Typically, Artificial Intelligence models need lots of training on lots of data in order to produce accurate translations, a fact that is hindered by the lack of sufficient data in low-resource datasets. NMT models are able to translate idiomatic expressions taking context into account by mimicking the cognitive functions of the human brain, weighing options before coming to a decision. These models still have issues with low-resource languages (see Figure 3) which can be improved with the implementation of attention algorithms. However, as with any form of A.I. Neural Machine Translation is biased.

GOALS

The goal of this experiment is to improve the performance of the Neural Machine Translation without further training. To find an attention mechanism that improves a model specifically for low-resource languages and to help make Neural Machine Translation accessible to different sociocultural identities, particularly those with low-resource datasets. In the process, it is also a goal to identify any possible overfitting in the results

HYPOTHESIS

The performance of the Neural Machine Translation model will be the most accurate with the implementation of the Transformer with the Multi-head attention mechanism because it will additionally incorporate the Scaled-dot product attention mechanism, but not be exaggerated in the number of mechanisms working at once.

This is based on the rationale above as using too many mechanisms at once may lead to overfitting while using only one mechanism may lead to issues in context. Therefore, Multi-head attention will be the most accurate.

METHODOLOGY

Model

OpenNMT-py, an open source repository by the Harvard NLP group, was used. The installation of Python 3.8 and PyTorch was required, so the project was done on a Linux based operating system, Ubuntu, on the command line. This operating system was the most readily available and robust. Working in the terminal provided more access in updating the GPU driver and version of Python on the computer. The NMT model was from Hugging Face Hub, an open source Machine Learning library. The model was pre-trained to account for costs and lack of cloud computing space. The chosen language pair was English-German with a low-resource dataset in order to mimic low-resource language pairs.

Advanced Techniques

Three different techniques for the models were used; no transformers implemented, Scaled-dot Product Attention Mechanism implemented, and Multi-head Attention Mechanism implemented. Scaled-dot Product Attention Mechanism calculates the dot product between query and key vectors although it can struggle with non-linear relationships in the data. It decides which parts of a phrase are more important through attention scores and compares the two sets of lists to find how similar they are, then weighs the words to place a numerical value of their importance for the model to understand. This mechanism works similarly to a person working with a group of people, each person representing a word or token, that pay attention to each other to ensure coordination in their work.

Multi-Head Attention Mechanism employs multiple parallel attention heads, capturing diverse patterns in the data more effectively, although it is most suitable for tasks requiring long-range dependencies. It breaks up input text and sends it to multiple heads of the model, resulting in an output consisting of the concatenated inputs. In the same scenario of working with a group of people, the Multi-Head Attention would be if all give their personal input about the same task from different perspectives.

These techniques were implemented through Shell scripts.

A Cross Attention Mechanism utilizes a separate embedding sequence in order to incorporate relevant input information for the output sequence. In the same scenario of working with a group of people, it is as if each member specializes in different areas and breaks the task up to focus on the specific details they specialize in.

Evaluation Metrics

Two evaluation metrics were used. BLEU (Bilingual Evaluation Understudy) is the most common metric for evaluating machine translations. It measures the precision of overlapping n-grams between the generated and reference translations, providing a score that reflects the similarity in terms of word sequences (See Figures 4 & 5). Compares input text with output text, Dividing the number of predicted words with the number of output words to generate a score between 0 and 1, with 1 being a perfect translation.

$$p_n = \frac{\displaystyle\sum_{C \in \{Candidates\}} \sum_{n\text{-}gram \in C} Count_{clip}(n\text{-}gram)}{\displaystyle\sum_{C' \in \{Candidates\}} \sum_{n\text{-}gram' \in C'} Count(n\text{-}gram')}.$$

Formula for Modified Precision from BLEU paper

$$\mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases}.$$

Then,

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right).$$

Formula for BP and BLEU from BLEU paper

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Figure 4. Figure 5.

PLP (Perplexity) metric used to evaluate the performance of language models by measuring how well they predict a given sequence of words. A PLP score nearer 0 indicates that the model is better at predicting the next word in a sequence. Measures the uncertainty the model has in the translation and how "surprised" the model is with the given input. The scores are negative, with a score nearer to 0 indicating the model is better at predicting the sequence of words.

DESIGN & PROCEDURE

1. Used a Linux based operating system, Ubuntu, so that Python 3.8 and Pytorch could be installed.
2. Downloaded four different models from Hugging Face Hub, an open source machine learning library: a non-Transformer model without attention mechanisms, a Transformer model with Scaled-dot product attention, a Transformer model with Multi-head product attention, and a Transformer model that uses Cross-head attention. The use of four different models prevented the results from being skewed due to the model learning from the previous tests. Used models trained on the same dataset.

3. Fine-tuned the models using the “train.py” command script that imported components from the “models” directory from OpenNMT
4. Obtained three different text inputs from three different sources of literature.
5. Divided into difficulty categories based on the Lexile metric the library that carries the texts used
6. Tokenized the inputs in a shell script
7. Ran the program using the command line “python3 translate.py –[model name] –src [script name] –replace_unk –verbose”
8. Ran the evaluation metrics using the “evaluate.py” command script
9. Recorded test input/output as well as PLP score and BLEU score
10. Repeated steps 6-9 using each text input
11. Changed command line to utilize next model “python3 translate.py –[next model name] –src [next script name] –replace_unk –verbose”
12. Repeated steps 10-11 using new model and test inputs
13. Collected data from each test
14. Ran bilingual evaluation understudy (BLEU) algorithm and Perplexity score using “evaluate.py” command script
15. Recorded BLEU and Perplexity score (PLP) in table
16. Repeated steps 13-15 for each test and model
17. Graphed and compared data
18. Looked for upward trends in both PLP and BLEU to indicate improvement

RESULTS

Cross-head attention mechanisms had the highest positive trend in both BLEU and PLP scores.

Model

The tests ran successfully with the command line “ python3 translate.py –model ende-large-withoutBT.pt –src x.txt –replace_unk –verbose “. With “python3” being the version of python the computer ran, “translate.py” being an installed program from OpenNMT-py, “--model” specifying the installed model being used, “--src” indicating the script containing the input text to be translated, and “--replace_unk” and “--verbose” setting parameters for how the translation will be done. There were issues in formatting the command with incorrect paths, which was resolved by moving the directory where the model was. Without implementing any advanced techniques, the model was case sensitive and marked capital “i” as an unknown word, leading to a higher perplexity score of -0.4553. When the text was properly tokenized the score went up to -0.2256.

Reading Level

Each technique was tested three times with varying levels of text difficulty: Elementary, Intermediate, and Advanced (See Test Transcript). The run time for the Scaled

Dot-Product mechanism was slightly longer than for the Multi head mechanism and the Cross mechanism.

ANALYSIS

The tests without any advanced technique performed poorly with the perplexity and BLEU score. There was a significant improvement with all the mechanisms, but slightly better with the Cross mechanism. All of the tests struggled in drastically improving scores with the advanced text. The perplexity score improved with each test, disregarding the text level which alters the score, showing how the model learned from each test without further training. Overfitting did not occur with the Cross-head attention mechanism, as could be seen with the increase in score in comparison to the Scaled-Dot Product and Multi-head.

PURPOSE

The purpose of this experiment was to improve Neural Machine Translation on low-resource datasets through attention mechanisms. Improve accessibility of translation using Artificial Intelligence and match the cognitive interpretations the human brain has towards linguistics in order to better translate for high-resource and low-resource languages alike.

NMT allows for translation that is accessible, accurate, and context-sensitive. This chips away at language barriers and facilitates multilingual communication to anyone with access to technology. By improving NMT models through testing of advanced techniques and further training, these models can become more equipped to handle input variations and become faster in training and interference. This could reduce maintenance on the model and eventually contribute to self-supervised or unsupervised learning for these models. Improving NMT can expand the benefits of machine learning in translation to languages that have low resources, although training models with small datasets still remains a challenge. This would promote language diversity and equity in the digital sphere.

FUTURE RESEARCH

NMT is a current and ever-evolving area with potential to improve in multiple aspects. This particular research could be made more accurate through using a dataset and training the model completely from scratch. Furthermore, using additional language pairs apart from English-German could make the research more extensive. Additional hours of training between tests could highlight the full benefits of the techniques used. Researching more advanced techniques for NMT apart from Attention Mechanisms could be beneficial in improving the models. Efforts to reduce bias in the model could be made by researching the data the model was trained on, keeping in mind cultural biases. Research through different forms of NMT models such as Speech-to-Text or Speech-to-Speech, instead of only Text-to-Text, could contribute to current advancements and be more applicable in daily life.

CONCLUSION

Contrary to the hypothesis, Cross attention mechanism improved the translation on the small dataset the best, as demonstrated by the BLEU and PLP scores. The implementation of attention mechanisms improved the translation, to varying degrees depending on the type of mechanism. These results have the potential to help in creating machine translation models for languages with low-resource language datasets. The results are constrained by the specific, high-resource English-German dataset that the model was originally trained on and could not accurately predict results for any model. The results could be better assessed with additional evaluation metrics such as a human evaluator.

REFERENCES

Bahdanau, Dzmitry, et al. "Neural Machine Translation by Jointly Learning to Align and Translate." arXiv preprint arXiv:1409.0473, 2014, https://arxiv.org/abs/1409.0473.

Burchardt, Aljoscha, et al. "Self-Attention with Cross-Attention for Transformer Models." arXiv preprint arXiv:2106.05786, 2021, https://arxiv.org/pdf/2106.05786.pdf.

Haris, Muhammad, et al. "Risks in Neural Machine Translation." ResearchGate, 2020, https://www.researchgate.net/publication/343737383_Risks_in_neural_machine_translation.

"OpenNMT." Hugging Face, https://huggingface.co/models?other=opennmt.

"OpenNMT." OpenNMT, https://opennmt.net/.

Košar, Václav. "Transformers: Self-Attention Mechanism Simplified." Václav Košar, https://vaclavkosar.com/ml/transformers-self-attention-mechanism-simplified.

Luong, Minh-Thang, et al. "Neural Machine Translation (seq2seq) Tutorial." Stanford University, 2017, https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1214/reports/final_reports/report096.pdf.

Luong, Thang, et al. "Attention Is All You Need." arXiv preprint arXiv:1706.03762, 2017, https://arxiv.org/abs/1706.03762.

Luong, Thang, et al. "Effective Approaches to Attention-based Neural Machine Translation." arXiv preprint arXiv:1508.04025, 2015, https://arxiv.org/abs/1508.04025.

Shen, Yufei. "Understanding and Coding Self-Attention." Sebastian Raschka, https://magazine.sebastianraschka.com/p/understanding-and-coding-self-attention#:~:text=What%20is%20cross%2Dattention%2C%20and,combine%20two%20different%20input%20sequences.